

Looking inside

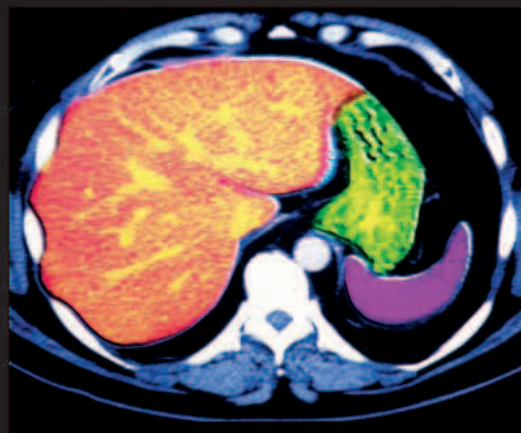
From the outside, the body rarely gives any clues about what is going on inside. One way of discovering how the body works is to use some of the astonishing medical technology that allows doctors to look inside a person without causing any harm. This also helps them spot any hidden injury or signs of disease.

X-RAY

When X-ray imaging was discovered in the late 1800s, doctors were able, for the first time, to see inside the living body without having to cut it open. This body imaging technique fires X-rays—a form of radiation—through a particular part of the body onto a photographic plate. Hard, dense parts of the body, such as bones, show up clearly since they absorb the rays. Softer body tissues are not visible since the X-rays pass right through them.

CT SCAN

In a CT (computed tomography) scan, beams of X-rays are sent through the body and are then analyzed by computer. This turns scans into images that show “slices” through the body. These slices can be built up into 3-D pictures to give an all-around view of an organ. CT scans are more revealing than ordinary X-rays because they show the soft tissues in greater detail.



▲ **CROSS-SECTION** This CT scan shows a slice through the abdomen, revealing the liver (orange), stomach (green), and spleen (pink).

Hard tissues
such as bone show
up as white areas.

Softer internal
organs show up as
gray or black areas
in an X-ray.

FAST FACTS

- X-rays were once used in shoe stores to examine how well the new shoes fitted each customer's feet.
- Patients undergoing an MRI scan have to remove any metal objects, such as jewelry, from their body because the scanner contains very strong magnets.
- Ultrasound is used in medicine to clean teeth, break up kidney stones, and remove cataracts from the eyes.